

Table 3. **The core: 80 works with a calibration term in the model’s own training loss or reward (S4)**, grouped by what the training signal scores; each work was assigned to one family by reading it. T1, the model is fitted to confidence targets or a calibration loss or regulariser, without a reward; T2, the reward model, a separate model in the training loop, is calibrated or made uncertainty-aware; T3, a reward or preference scores the stated confidence against the outcome, including proper scoring rules and ranking-based (AUC) confidence rewards, which target discrimination as well as calibration; T4, the signal targets the action: abstaining or refusing when unsure (supervised or rewarded); T5, the confidence reward is given per reasoning step or over meaning-equivalent answers. Family sizes: T1 23, T2 5, T3 31, T4 17, T5 4. The 5 theory and analysis works that follow are not counted in the core. * marks the 14 general-domain calibration-training works of 2025 and 2026 of which no prior survey cites more than one. Yr: year of the first public version (a citation may show a later venue year); Contract: the form of the output the confidence is attached to: typed (a probability per declared option), set (a prediction set), constrained (one output under a grammar or schema), score (one confidence number per answer, stated, read off or estimated by a separate model) or free-text (free text or a label with no per-answer probability; any confidence is a verbal hedge, an abstention, or agreement across sampled answers); Action: what an explicit rule or a trained behaviour does with the confidence when it is low (abstain; hand off = pass the case to another model or a human, which the papers call routing, cascading, deferral or escalation; retrieve = trigger retrieval; filter = remove, rewrite or revise low-confidence claims, or keep the most confident of several sampled answers; adapt compute = keep sampling, reasoning, querying or exploring, or spend a larger compute budget (the same rule stops early when the confidence is high); set = return a prediction set; none = no rule or trained behaviour acts on the confidence, including a selective-prediction curve reported only as an evaluation); Guar.: a stated distribution-free guarantee; Metric: the calibration or confidence metric reported in the main text (ECE, Brier, NLL, AUROC, selective = selective risk or accuracy at a coverage, coverage = achieved coverage of a prediction set or achieved risk under a controlled threshold, multiple = more than one of these (a reliability diagram counts with ECE; PRR with AUROC), other = a different metric, none = no confidence metric). ? = not determinable from the paper.

Work	Yr	Contribution	Contract	Action	Guar.	Metric
T1 supervised calibration objectives (23)						
Jiang et al. [2021]	2020	Finds T5, BART and GPT-2 QA probabilities are poorly calibrated and compares fine-tuning, temperature scaling and paraphrase fixes	typed	none	no	ECE
Kong et al. [2020]	2020	on- and off-manifold regularisers added to fine-tuning loss for in- and out-of-distribution calibration	typed	none	no	ECE
Lin et al. [2022]	2022	Fine-tunes GPT-3 to state calibrated numeric or word-level confidence without logits and introduces the CalibratedMath suite	score	none	no	multiple
Chaudhry et al. [2024]	2024	supervised fine-tuning on uncertainty-augmented answers to emit calibrated linguistic hedges	free-text	none	no	multiple
Chuang et al. [2024]	2024	Model emits a learned confidence token with each answer and low confidence triggers routing to a larger LLM	score	hand off	no	multiple
Han et al. [2024]	2024	LePe: fine-tunes on records of past answer correctness so the model learns to express its own confidence	score	none	no	ECE
Kapoor et al. [2024]	2024	calibration tuning on a small graded dataset of correct and incorrect answers yields generalising uncertainty estimates	score	none	no	multiple
Krishnan et al. [2024]	2024	uncertainty-aware causal LM loss grounded in decision theory raises uncertainty on wrong tokens	score	none	no	multiple

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Work	Yr	Contribution	Contract	Action	Guar.	Metric
Liu et al. [2024c]	2024	self-training scheme in which the LLM learns to express uncertainty from its own consistency signals	score	retrieve	no	AUROC
Niu et al. [2024]	2024	UQ4CT: calibration loss over mixture-of-LoRA-experts functional space during PEFT	typed	none	no	ECE
Wang et al. [2024e]	2024	BLoB: jointly learns mean and covariance of LoRA weights by backpropagation for calibrated fine-tuning	typed	none	no	multiple
Zhang et al. [2024d]	2024	self-evaluation calibration tuning (SK-Tuning) then DPO on self-annotated factuality preferences	score	none	no	multiple
Eikema et al. [2025]	2025	Faithful Uncertainty Tuning: SFT on self-samples augmented with hedges matched to sample consistency	free-text	none	no	other
Hager et al. [2025]	2025	uncertainty distillation: SFT on verbalised probabilities derived from Monte Carlo semantic sampling	score	none	no	multiple
Huang et al. [2025a]	2025	Self-Calibration: distills self-consistency confidence into a single-pass confidence for efficient test-time scaling	score	adapt compute	no	multiple
Jang et al. [2025]	2025	SFT on scalar confidence labels alone elicits self-verification in chain-of-thought	score	adapt compute	no	multiple
Li et al. [2025d]*	2025	ConfTuner: tokenised Brier score loss (a proper scoring rule) for verbalised confidence fine-tuning	score	hand off	no	multiple
Steyvers et al. [2025a]	2025	tests cross-domain transfer of supervised fine-tuning for numeric confidence and pairwise confidence comparison	score	none	no	multiple
Xiao et al. [2025]	2025	calibration-aware fine-tuning plus EM-based ECE regulariser to restore calibration lost in preference alignment	typed	none	no	ECE
Zhang et al. [2025c]	2025	Direct Confidence Alignment: DPO aligns verbalised confidence with internal token-probability confidence	score	none	no	other
Zhang et al. [2025b]	2025	GrACE: fine-tunes a special confidence token whose hidden-state similarity gives calibrated confidence	score	adapt compute	no	multiple
Guo et al. [2026]	2026	post-trains end-of-reasoning verbalised confidence and an in-reasoning <uncertain> marker used to trigger RAG	score	retrieve	no	multiple
Xie et al. [2026]	2026	post-RL SFT with self-distillation restores anchor-token confidence reliability degraded by GRPO and DPO	typed	retrieve	no	multiple
T2 reward-model calibration (5)						
Leng et al. [2025]	2024	PPO-M calibrated reward modelling with confidence-augmented pairs and PPO-C reward adjustment by running mean	score	none	no	multiple
Lou et al. [2024]	2024	URM: reward model outputs mean and variance of attribute rewards to expose reward uncertainty	score	none	no	selective
Park et al. [2025a]	2025	quantile-regression calibration of process reward models used for instance-adaptive compute	score	adapt compute	no	multiple

Table 3, continued

Work	Yr	Contribution	Contract	Action	Guar.	Metric
Ma et al. [2026a]	2026	conditional optimal transport over PRM hidden states yields calibrated success quantiles	score	adapt compute	no	multiple
Pan et al. [2026a]	2026	UARM: quantile conformal uncertainty for reward models plus variance-reweighted GRPO advantages	set	none	yes	selective
T3 calibration rewards on the stated confidence (31)						
Band et al. [2024]	2024	linguistic calibration: SFT then RL rewarding generations that let a simulated reader make calibrated forecasts	free-text	none	no	ECE
Stengel-Eskin et al. [2024]	2024	listener-aware DPO where preferences reward answers a simulated listener accepts only when correct	free-text	abstain	no	multiple
Tao et al. [2024]	2024	CONQORD: PPO with quality reward plus order-preserving reward aligning verbalised confidence with response quality	score	retrieve	no	ECE
Xu et al. [2024a]	2024	SaySelf: SFT on self-reflective rationales then RL with a reward penalising overconfident wrong answers	score	none	no	ECE
Bani-Harouni et al. [2026b]*	2025	Rewarding Doubt: log scoring rule reward on verbalised confidence	score	none	no	multiple
Damani et al. [2025]*	2025	RLCR: binary correctness reward plus Brier score reward on reasoned confidence	score	none	no	multiple
Kale and Dhami [2025]	2025	KnowRL: introspection plus consensus-based self-reward to strengthen feasibility self-knowledge	free-text	none	no	other
Liu et al. [2025b]	2025	C2GSPG: group sequence policy gradient with a cross-entropy regulariser aligning sequence confidence with reward	score	none	no	multiple
Turtel et al. [2025]	2025	outcome-based RLVR on resolved prediction-market events that improves forecast calibration	score	none	no	multiple
van Niekerk et al. [2025]	2025	RLSF: preference optimisation using the frozen model own answer-span confidence as self-feedback reward	score	none	no	ECE
Wu et al. [2025a]*	2025	behaviourally calibrated RL with strictly proper scoring rule rewards enabling abstention or claim flagging	score	abstain	no	multiple
Yin et al. [2026]	2025	Deliberative Searcher: search agent RL maximising accuracy under a soft reliability (confidence) constraint	score	none	no	other
Zhang et al. [2025e]	2025	LoVeC: GRPO and DPO with calibration reward for sentence-level confidence tags in long-form output	score	none	no	multiple
Zou et al. [2025]	2025	GRPO with composite reward penalising unjustified high confidence and token-entropy spikes	score	none	no	multiple
Bani-Harouni et al. [2026a]	2026	ConRad: RL with calibration reward for report- and sentence-level verbalised confidence in radiology LVLms	score	none	no	multiple
Cui et al. [2026]	2026	RAC: ranking-aware group loss and clean-corrupted pairwise confidence loss inside multimodal RL	score	none	no	multiple

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Li et al. [2026d]	2026	CoCA: confidence-first GRPO with segmented Brier-style confidence reward and answer reward	score	none	no	multiple
Li et al. [2026a]	2026	ORCE: rank-based RL reward aligning post-hoc verbalised confidence order with sampled correctness surrogate	score	none	no	multiple
Liu et al. [2026a]	2026	RLMF: preference optimisation whose completion rankings use metacognitive feedback on self-judgement quality	score	none	no	Brier
Liu and Wang [2026]	2026	CURE: claim-level confidence reasoning trained with DPO confidence correction and RL	score	filter	no	multiple
Ma et al. [2026b]*	2026	DCPO: decouples reasoning and calibration objectives with separate advantages to avoid gradient conflict	score	none	no	multiple
Singh et al. [2026b]	2026	studies verifiable Brier rewards for aleatoric forecasting (NFL win probability) and why per-outcome reward fails	score	none	no	multiple
Tan et al. [2026]*	2026	characterises non-hackable confidence reward schemes (Brier, log hybrids) against confidence reward hacking	score	none	no	multiple
Turtel et al. [2026]	2026	compares five proper scoring rules (Brier, log, others) as terminal RL rewards for LLM forecasters	score	none	no	multiple
Wang et al. [2026a]*	2026	CAPO: logistic AUC surrogate loss for uncertainty-aware advantage estimation in GRPO	score	none	no	multiple
Xiao et al. [2026]	2026	VL-Calibration: RL rewarding separate visual and reasoning confidence in LVLMS	score	none	no	multiple
Xu et al. [2026a]	2026	DualStake: margin-clipped confidence-dependent stake rewards for evidence and answer confidence in deep research agents	score	none	no	multiple
Xuan et al. [2026]	2026	RL fine-tuning jointly rewarding task accuracy and verbalised calibration in tool-use agents	score	none	no	multiple
Yaldiz et al. [2026]*	2026	calibration-aware RLVR that directly adjusts decision-token probabilities to curb overconfidence	typed	none	no	ECE
Yang et al. [2026b]	2026	C3RL: correctness reward plus calibration reward plus dataset reference-accuracy reward for adaptive test-time scaling	score	adapt compute	no	multiple
Zhu [2026]	2026	RefGRPO: free calibration bonus contrasting agent reflection score with actual environment outcome	score	filter	no	selective
T4 abstention and refusal training (17)						
Yang et al. [2024b]	2023	alignment for honesty: fine-tuning to answer known questions and say I do not know otherwise, with honesty metrics	free-text	abstain	no	other
Zhang et al. [2023a]	2023	R-Tuning: builds refusal-aware SFT data by appending uncertainty expressions to questions the model gets wrong	score	abstain	no	other

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Chen et al. [2025a]	2024	CoKE: probes knowledge boundary via internal confidence then trains the model to express it	free-text	abstain	no	other
Cheng et al. [2024]	2024	Idk dataset built per model and SFT plus preference optimisation to refuse unknown questions	free-text	abstain	no	other
Kang et al. [2025b]	2024	shows unfamiliar finetuning examples shape hallucination and proposes more reliable reward models for RL factuality finetuning	free-text	abstain	no	other
Li et al. [2025c]	2024	US-Tuning: two-stage instruction tuning for knowledge-boundary awareness then instruction adherence	free-text	abstain	no	other
Liang et al. [2024a]	2024	RLKF: knowledge-state preference data from DreamCatcher used as reward for honesty-oriented RL	free-text	abstain	no	other
Xu et al. [2024b]	2024	RLKF with a reliability reward model trained on knowledge-feedback preferences to refuse out-of-knowledge questions	free-text	abstain	no	other
Xue et al. [2025]	2024	UAlign: uncertainty estimates (confidence, semantic entropy) as features for SFT and PPO factuality alignment	free-text	abstain	no	AUROC
Mohamadi et al. [2025]*	2025	Reinforced Hesitation: ternary RLVR reward (+1, 0, -lambda) with penalty-dependent abstention frontier	free-text	abstain	no	other
Song et al. [2025]	2025	shows reinforcement finetuning erodes refusal and restores it by mixing synthetic unanswerable math	free-text	abstain	no	other
Wei et al. [2025]*	2025	TruthRL: ternary reward (correct +1, abstain 0, hallucination -1) under GRPO	free-text	abstain	no	other
Che et al. [2026]	2026	proves error-penalised abstention rewards can collapse KL-anchored RL to refuse-all and proposes a repair	score	abstain	no	Brier
Gao et al. [2026]*	2026	KARL: knowledge-boundary-aware reward from within-group statistics plus two-stage RL	free-text	abstain	no	other
Pan et al. [2026b]	2026	TIAR: trajectory-informed reweighting of the abstention advantage within GRPO ternary reward	free-text	abstain	no	other
Song et al. [2026]	2026	Visual-Idk dataset and SFT plus preference optimisation so VLMs refuse unknown visual facts	free-text	abstain	no	other
Zhai et al. [2026]*	2026	Abstain-R1: clarification-aware RLVR reward for explicit abstention plus post-refusal clarification	free-text	abstain	no	other
T5 process- and meaning-level confidence rewards (4)						
An and Xu [2025]	2025	FiSCoRe: fine-grained sample-specific semantic confidence reward guiding abstention in GRPO	free-text	abstain	no	other
He et al. [2025]	2025	MMBoundary: SFT on step-level self-rewarded confidence then RL with step calibration rewards for MLLMs	free-text	none	no	ECE
Wang et al. [2026d]*	2026	RLCM: margin-enhanced process reward on confidence at intermediate reasoning budgets plus Brier term	score	abstain	no	multiple

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Yu et al. [2026b]*	2026	CSR: correctness reward plus semantic calibration reward on agreement among rollouts, no verbal confidence	free-text	none	no	multiple
Theory and analysis of training-stage calibration (not counted in the core) (5)						
Blasiok et al. [2023]	2023	relates proper-loss local optimality to smooth calibration, explaining when training yields calibration	?	none	no	none
Kalai and Vempala [2024]	2023	proves calibrated LMs must hallucinate arbitrary facts at a rate near the Good-Turing fraction of facts seen once	free-text	none	no	none
Bereket and Leskovec [2025]	2025	shows GRPO group-std normalisation causes overconfidence on stochastic outcomes; PPO and RLOO stay calibrated	score	none	no	multiple
Kalai et al. [2025]	2025	argues binary-graded training and evaluation reward guessing and calls for scoring that credits admitting uncertainty	free-text	none	no	none
Nakkiran et al. [2025]	2025	theory that semantic calibration emerges from next-token loss and breaks under RL tuning and CoT	free-text	none	no	ECE